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LLM+RAG+Agent: Create an efficient and accurate text data labeling system

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***Abstract*—With the rapid development of generative artificial intelligence, it has shown great potential in many fields, and the construction of text data labeling system has also ushered in new opportunities for change. This paper conducts in-depth research on building an efficient and accurate text data labeling system, focusing on the innovative applications of large language models, RAG, and agents. To overcome the key problem of high-precision intelligent annotation, an intelligent annotation method of vector semantic matching driven by large model was innovatively proposed. Firstly, the problems to be solved by smart labels are analyzed in detail, and then they are reasonably modeled, and then the abstract is extracted with the help of large model driven, and the annotation method design is completed by skillfully matching with the label content system. At the same time, agents play an indispensable auxiliary role in the optimization process of RAG technology search. By deeply analyzing the structure, topic and other characteristics of the input text, the agent can automatically generate accurate search keywords or phrases. After rigorous analysis of the experimental results, the accuracy of the method is verified to be more than 95%, and the application scenarios of the method are described.**

Keywords-Large language models; RAG; agents; smart labeling system

I. INTRODUCTION

In the current explosion of digital information, the development of generative AI is changing rapidly, profoundly changing the operation mode of many fields. Massive amounts of text data are emerging, and the efficient management and utilization of these data has become a key problem to be solved in various industries. As an effective means of data organization and understanding, the importance of text data labeling system is becoming more and more prominent. Accurate labels can quickly locate and classify data, helping enterprises and researchers extract key value from complex information, and play an indispensable role in information retrieval, data analysis, and knowledge graph construction. In the past, text labeling methods mostly relied on manual annotation or automatic annotation based on rules-based and simple machine learning models. Although manual annotation can ensure a certain degree of accuracy, it is time-consuming, laborious, inefficient, and costly, and it is difficult to cope with the annotation needs of large-scale data. The rule-based annotation method is inflexible and often unable to cope with the complex and changeable semantics and structure of the text. However, the simple machine learning model is limited by the quality of feature engineering, lacks generalization ability, and performs poorly when processing texts in new fields and new topics, which cannot meet the dual requirements of high efficiency and accuracy of text data processing. To break through the dilemma of traditional text labeling, this paper innovatively proposes a solution that combines large language model, Retrieval Augmented Generation (RAG) technology and agents. With its powerful language understanding and generation capabilities, large language models can deeply dissect text semantics and provide strong support for label extraction. RAG technology makes up for the shortcomings of large language models in knowledge updating and domain-specific information acquisition by accurately obtaining relevant knowledge in the retrieval process, and enhances the accuracy and reliability of label generation. The agent can

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automatically optimize the search strategy according to the text features, further improving the efficiency and accuracy of the entire system. This multi-technology integration method is expected to build a new, efficient and accurate text data labeling system, and open up a new path to solve the problem of text data management.

In this paper, the label content system is designed, the intelligent label annotation model is constructed, the semantic understanding and content generation capabilities of large language model technology are used to extract abstract unstructured text data, and the RAGGING is assisted by agents, and the retrieval strategy is automatically optimized according to text features, so as to further improve the efficiency and accuracy of the whole system. It is labeled by semantic matching with the tag library after vectorization. Large language models are prepared in the data cleansing and preprocessing stages, the model architecture is selected, and then pre-trained by unsupervised learning. Fine-tune instructions on specific tasks to suit the needs of the task. Finally, the optimization and evaluation of the model were carried out. Figure 1 shows the technical process.

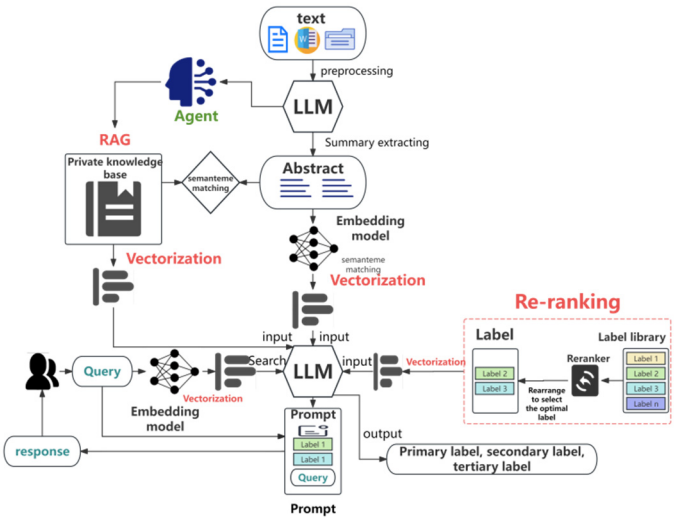

Figure 1. Design diagram of the intelligent labeling system

II. Related Work

A. Large language models

Since its inception, large language models (LLMs) have set off a wave of revolution in the field of natural language processing with their excellent language understanding and generation capabilities [1]. Early language models are limited by the number of parameters and data scale, and their semantic understanding and generalization capabilities are weak. With the development of deep learning, the proposal of the Transformer architecture has become a key turning point, and its unique self-attention mechanism allows the model to effectively capture long-distance dependencies and significantly improve the understanding of context [2].

B. RAG

Retrieval Enhanced Generation (RAG) technology aims to combine information retrieval with text generation to make up for the shortcomings of large language models in knowledge updating and knowledge acquisition in specific domains. Early text generation models relied on pre-trained knowledge and were difficult to obtain the latest information or domain-specific expertise in real time, resulting in limitations in the generated content [3]. The advent of RAG technology has enhanced the reliability and relevance of generated content by retrieving an external knowledge base during the generation process to provide models with real-time, accurate information [4].

C. Agent

Driven by advances in large language models (LLMs), LLM-based agents have demonstrated powerful capabilities. Based on LLMs, these agents combine the autonomy, perception, decision-making, and action capabilities of agents, with powerful natural language processing and comprehensive knowledge to handle a wide range of natural language tasks, excel in zero-shot or few-shot learning, and enable organic human-computer interaction. On March 31, 2025, the paper "Advances and Challenges in Foundation Agents", jointly released by 19 institutions including Meta, Stanford, and Google DeepMind, proposed an agent development framework based on large language models [5]. The framework transforms the biological mechanism of human cognition into computable modules, realizes autonomous decision-making ability through reinforcement learning, multimodal interaction and other technologies, redefines "agent" as a complex system with "brain-like" cognitive ability, and proposes a six-dimensional architecture of "cognition-perception-memory-world model-emotion-action". Deep learning-based annotation models have developed rapidly, such as BERT-based entity recognition [6] and LSTM-based sentiment analysis [7]. These models can automatically learn the features of the text, improving the efficiency and accuracy of annotation. For example, the Text-Tag Self-Distillation (TTD) method is proposed to address this challenge, in which all image-related labels are extracted from the text based on its similarity to the nearest pixel [8].

D. Text Data Labeling

The previous research has made preliminary attempts on large language models, RAG and agent fusion. Some of the work uses large language models and RAG for text generation and annotation, but there are shortcomings in the agent's intelligent scheduling of the entire process and the dynamic adjustment of strategies according to the annotation results. Some agent-assisted text processing systems do not make full use of the powerful language understanding and RAG knowledge retrieval advantages of large language models, and the depth of collaboration between the three is insufficient, and an efficient and accurate text data labeling system has not been formed. In contrast, this paper aims to build a closely integrated and complementary system to improve the efficiency and accuracy of text data labeling in an all-round way.

III. Smart Labeling System

This chapter focuses on the core method design of the smart label system. Firstly, for the construction of the private knowledge base, a phased data processing strategy is adopted to realize the standardized processing of raw data through text preprocessing, data collection and filtering, multi-source data integration, unstructured data structured conversion, data

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cleaning and standardization. On this basis, the vector space model is used to vectorize the processed data to construct a private knowledge base with efficient retrieval ability.

In the process of label generation and optimization, combined with the natural language understanding ability of large language model (LLM), private data is effectively input into the large language model through contextual semantic parsing and prompt engineering technology. By assisting the process of abstract vectorization and label vector matching, the interpretability and dynamic update ability of label results are significantly improved, which lays a foundation for the accurate application and iterative optimization of subsequent labels. The specific method process design is shown in Figure 2.

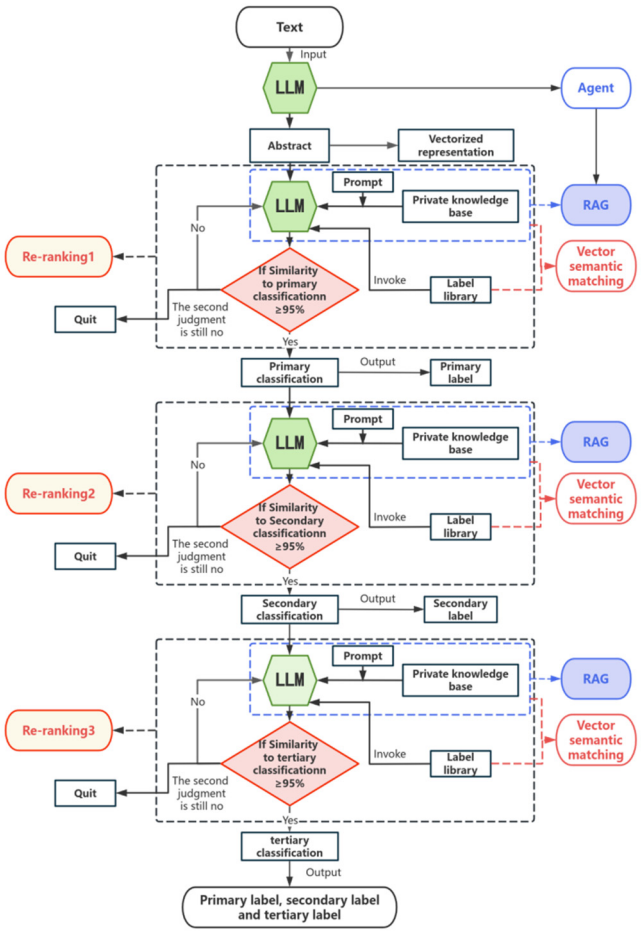

Figure 2. This diagram shows the information flow diagram of the intelligent labeling system.

(1) The private knowledge base retrieving

The formula based on the self-attention mechanism is as follows:

$$Attention(Q,K,V) = soft\max\left(\frac{QK^T}{\sqrt{d_k}}\right)V \qquad (1)$$

Among them, Q stands for Query, indicating the query of the element currently being processed for other elements. K stands for Key, indicating the key of each element in the dataset, and is used to determine which elements in the private knowledge base dataset are related to the current element. V stands for Value, representing the value of each element in the private knowledge base dataset, which contains the actual content of the selected (matched by keys) element. It is the dimension of the key vector. Taking the square root is to scale the dot product to prevent the vanishing gradient problem. Then, the attention score is standardized by applying the softmax function.

The agent generates a comprehensive score formula for keywords

Let the input text be T. The set of candidate keywords generated by the agent based on the text features is K={k_1, k $_2$, ... kn}. For each candidate keyword ki, the agent considers the following factors to calculate its score:

Semantic Relevance score Ssemantic(ki,T): Measures the keyword k; The degree of semantic relevance to the input text T can be obtained by calculating semantic similarity using a pre-trained language model.

Structural Importance score Sstructure(ki,T): Considering the text structure, for instance, the score is relatively high when keywords appear in important positions such as the title and the beginning of a paragraph.

Agent experience weight Wagent: It reflects the agent's performance and experience accumulation in previous retrieval tasks and is a constant ranging from 0 to 1.

The calculation formula for the comprehensive score S(ki) of the candidate keyword k is:

S (ki) = Wagent x [a x Ssemantic (ki, T) + beta * Sstructure (k, T)]

Among them, α and β are parameters for adjusting the relative weights of semantic relevance and structural importance, and α+β=1. Finally, the key words with higher comprehensive scores are selected as the search keywords. It is worth noting that to verify the effectiveness of the adaptive adjustment strategy for semantic weight α and structural weight β in keyword scoring formulas, the values of α and β can be adjusted in real-time based on the domain characteristics, structural complexity, and semantic density of the text. By conducting statistical tests (such as t-tests) to confirm significant differences, it can be demonstrated that this adaptive strategy is effective and feasible in achieving domain adaptation and optimizing feature weight allocation.

(2) Pre-training

During the pre-training process, vectorization representation is carried out first, which can be divided into two dimensions. On the one hand, the labels are vectorized. The existing labels in the label library can be vectorized by referring to the word vector method. By conducting pre-training on large language models, word vectors containing semantics can be learned. The Word vectors are embedded using the Word2Vec method (Word Embeddings), and the

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word embeddings are obtained through neural network training to capture the meanings and grammatical relationships between words. Transfer the pre-trained word vectors to the task to improve the subsequent effect. If the label vector is represented as W and C is the number of labels in the label library, then the average L of its label vector is:

$$L = \frac{1}{C} W^T \left(x_1 + x_2 + x_3 + ... + x_C \right) \tag{2}$$

Transforming the one-time V classification task of label classification into a logV binary classification task can reduce the training time. N represent the random probability noise, The positive example is σ, and the negative example is N^-, The context is represented by C, the minimized log-likelihood function of the word vector is $F(N)$, the formula is expressed as follows:

$$F(N) = \sum_{t=1}^{T} \sum_{N^+ \in c_t} \left(\log \sigma \left(N_t^T N^+ \right) \right) + \sum_{N^- \in c_t} \left(\log \sigma \left(-N_t^T N^- \right) \right) \tag{3}$$

On the other hand, unstructured text is represented by text vectorization. The TF-IDF method (Term frequency-Inverse Document Frequency Tern frequency-inverse Document Frequency) is adopted to vectorize the document by evaluating the importance of feature words in the document.

$$TF_{i,j} = \frac{n_{i,j}}{\sum_{u=1}^{u} n_{u,j}} \tag{4}$$

Among them, $TF_{i,j}$ represents the frequency of the word w_i in document d_j, $n_{i,j}$ is the occurrence number, and u is the number of different words that appear in document d_j.

$$IDF_i = \log \left(\frac{n_d}{df(d, w_i) + 1} \right) \tag{5}$$

At this point, n_d represents the number of documents, and $df(d, w_i)$ represents the number of documents containing feature words in the documents.

Therefore, the vector T_i of the text is expressed as:

$$T_i = TF_{i,j} \times IFT_i \tag{6}$$

(3) Design Prompt engineering

Set $X = [x_1, x_2, ..., x_n]$ as the summary text to be marked, $Y = [y_1, y_2 ..., y_n]$ as the target tag, and P as the prompt word for the project, totaling n text data.

$$P_\varepsilon \left(y | x, P \right) = \prod_{i=1}^{n} p_\varepsilon \left(y_i | x, P, y \le i \right) \tag{7}$$

Then, based on the task prompt, the embedding model is introduced, specifically the label entity embedding, to transform knowledge extraction into the labels of the embedding model and assign tasks.

$$I_{Label} = \{ el_1, el_2, ..., el_n \} \tag{8}$$

Among them, el represents the entity label.

(4) Calculate the semantic matching similarity of vectors

For the matching stage of label vectors and summary vectors in the label library, the cosine similarity is calculated to determine whether the vectors meet the classification requirements, thereby matching the labels.

$$similarity\left(L, TF_{i,j} \right) = \frac{L \bullet T}{\|L\| \times \|T\|} = \frac{\sum_{i=1}^{n} L_i T_i}{\sqrt{\sum_{i=1}^{n} L_i^2} \times \sqrt{\sum_{i=1}^{n} T_i^2}} \tag{9}$$

Determine the similarity. If the similarity is ≥95%, output. Otherwise, re-input the large model for secondary determination.

For the matching situation between the private knowledge base and the summary, the semantic matching method is adopted to conduct semantic similarity matching between the application description and the private knowledge base, and respectively transform them into semantic list patterns. P represents the semantic list of the private knowledge base, and A represents the semantic list of the generated summary. Determine whether the similarity meets the 95% threshold and output the label. The purpose of semantic matching is to facilitate search using tags. Therefore, the calculation formula of semantic similarity is:

$$\text{si}milarity\left(A, P \right) = \frac{\sum_{i=1}^{n} similarity\left(A_i, P_i \right)}{n} \tag{10}$$

(5) Data recall Re-ranking mechanism

In the process of vector semantic matching, the data recall mechanisms of Ranking and Re-Ranking are utilized. The Ranking is represented by the function $f_{ranking}$:

$$R = \left\{ \left(L_{pre_j}, R_j \right) \middle| L_{pre_j} \in L_{pre}, R_i = f_{ranking} \left(L_{pre_j}, u \right) \right\} \tag{11}$$

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Among them, L_{pre} is the pre-selected label, R is a set of matching pairs, u is the user's intention, and R_i represents the relevance score. In the reordering mechanism, in order to enhance the repeatability of the method, we explicitly choose the BM25 algorithm as function $f_{ranking}$ and Re-Ranking function:

$$Score(q,d) = \sum_{i=1}^{n} IDF(t_i) \times \frac{f(t_i,d) \times (k_1+1)}{f(t_i,d) + k_1 \times \left(1 - b + b \times \frac{|d|}{avgdl}\right)} \tag{12}$$

Here, $Score(q,d)$ is the relevance score between query q. and document d. The higher the score, the higher the relevance between document d and query q. Here, n is the number of terms contained in query q.

After sorting, the matching similarity calculation is carried out. The text vectors that do not meet the condition of similarity ≥95% need to be returned to the large model for reordering. At this time, a self needs to be separated from the previously sorted self for reordering. The reordering function is f_{re}. During the reordering process, L_{re} is adjusted according to the credibility score s_t related to each document item. The final output is expressed as O:

$$O = \left\{ \left(L_{re_t}, O_t \right) \middle| L_{re_t} \in L_{re}, O_t = f_{re}\left(L_{re_t}, s_t \right) \right\} \tag{12}$$

According to the requirements of the three-level labels, the calculation of this model needs to be trained in three rounds in sequence. The output result of the first round is the first-level label, the output result of the second round is the second-level label, and so on. The final output results are first-level labels, second-level labels, and third-level labels.

IV. Experimental Validation

In this experiment, the large language model GLM4-9B jointly released by Tsinghua University and Zhipu AI was selected in the method of "intelligent annotation of large language model and expert verification", and the LangChain development framework was used for the experiment. A total of 300 texts in three categories of law, medicine, and journalism were selected for the experiment. The experimental process is described as follows.

Firstly, the label content system is vectorized to form a label library, which is divided into news text, medical text, and legal text, as shown in Figures 3-5.

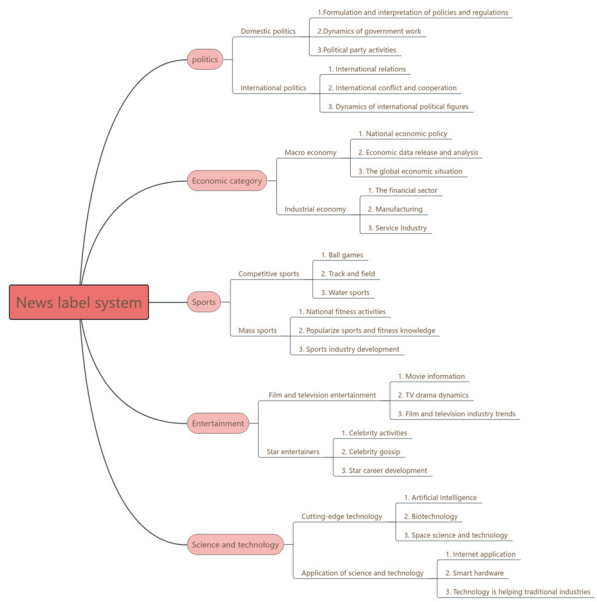

Figure 3. This is the news label system.

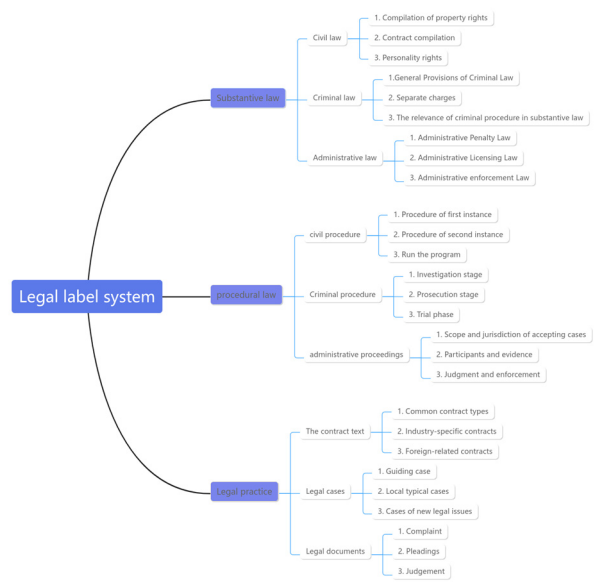

Figure 4. This is the legal label system.

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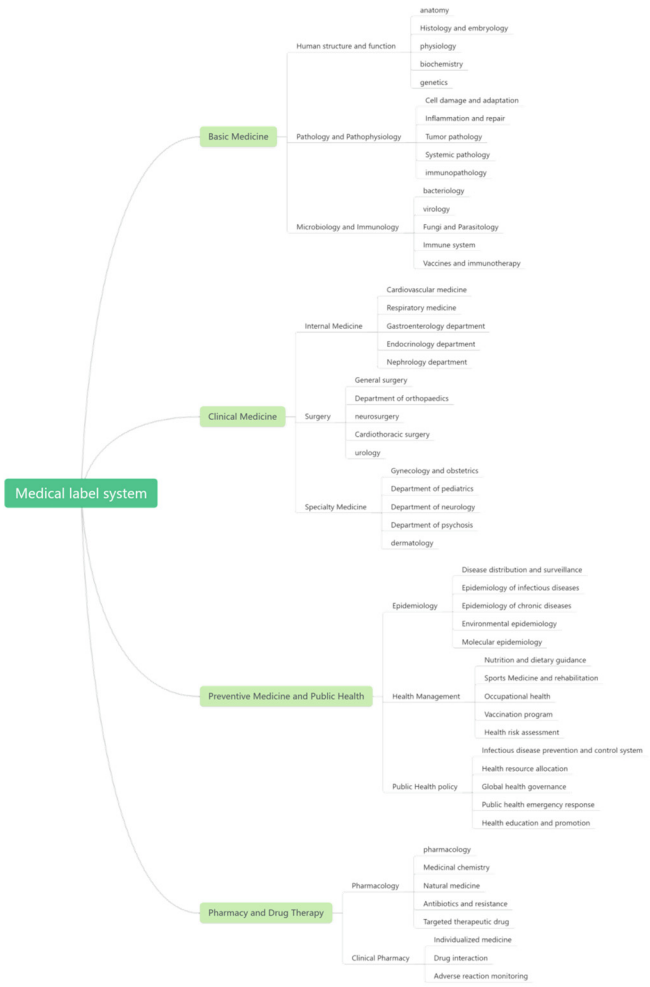

Figure 5. This is the medical label system.

Accuracy is a key metric for validating the performance of smart labeling. In this section, the performance evaluation of the method of "expert verification of large model annotation" is emphasized. A total of 300 documents were selected, according to the three categories of medicine, law, and news, but the number of copies was unequal to the input of the large model, firstly, the large language model and the agent were combined to classify the text, and if the medical text was retrieved and enhanced according to the medical label content system, the output label was output. Then, the 300 documents were verified by five experts, and the accuracy of intelligent annotation after verification by each expert was calculated, and the experimental results are shown in Figures 6-8.

Therefore, we performed the statistics according to the three categories of agent-assisted RAG post-classification.

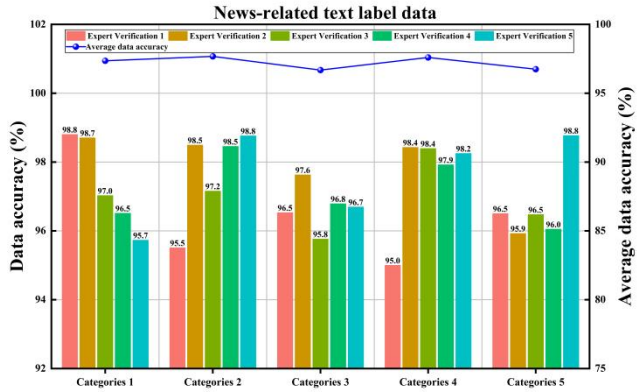

Figure 6. This is the legal label system

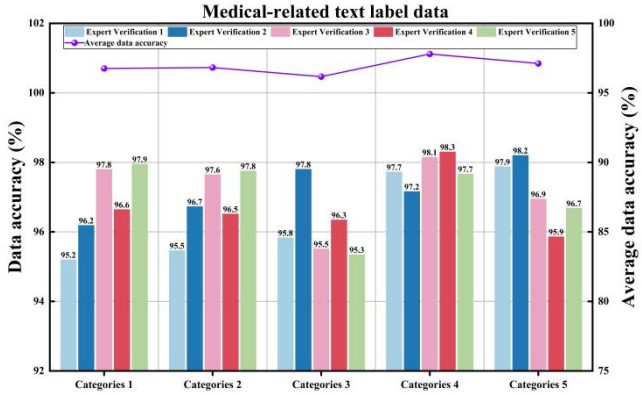

Figure 7. This is the medical label system.

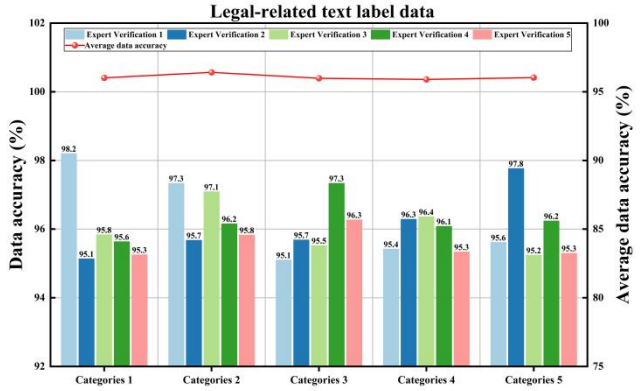

Figure 8. This is the legal label system.

Through the analysis of the following three charts ("News - related text label data", "Medical - related text label data" and the third figure not shown), it is clear that the accuracy of smart labeling has always been above 95% during the expert verification process of different types of text data.

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In terms of news text label data, the accuracy of each category of data fluctuates under the verification of different experts, but all of them are at a high level. For example, in the "Categories 1" category, Expert Verification 1 achieved an accuracy rate of 98.8%, and even the relatively low Expert Verification 5 achieved an accuracy rate of 95.7%. The same is true for medical text label data, such as Expert Verification 4 with an accuracy rate of 98.3% in the "Categories 4" category and 97.2% in the lowest Expert Verification 3.

These data strongly show that the method of creating an efficient and accurate text data labeling system based on large language model, RAG and agent technology has significant feasibility at the experimental level. It can stably and reliably achieve high-precision label annotation in text data processing in different fields, laying a solid foundation for efficient processing of text data and improving information management and analysis efficiency in practical application scenarios.

V. Application Scenarios

The text data labeling system built by the deep integration of large language models, RAG and agents has shown significant application value in many fields. In document processing scenarios, in the face of complex format documents such as contracts, financial reports, and academic papers, the system can not only parse the text, but also process embedded tables, formulas, and pictures, and build an enterprise-level knowledge base through data cleaning and vectorized storage, greatly reducing the reading threshold of professional documents, and supporting knowledge Q&A and content generation. In the field of information retrieval and knowledge Q&A, the system converts user questions into content vectors, accurately recalls knowledge base information, outputs reliable answers with the help of large language models, and marks the citation source, effectively avoiding the "hallucination" problem. In the data annotation scenario, the automatic classification and annotation of large-scale text data can be realized through custom prompts and learning strategies, which significantly accelerates the data preparation process of machine learning projects, and uses the cost estimation function to ensure that the budget is controllable. In terms of office process automation, relying on the decision-making capabilities of large language models and the knowledge support of RAG, agents decompose tasks into sub-steps, link databases, APIs and other tools to efficiently complete office tasks such as customer service response and cross-language translation, and comprehensively improve work efficiency.

VI. Conclusions

This paper focuses on utilizing large language models, RAG, and agents to create an efficient and accurate text data labeling system. To address the challenge of high-precision intelligent annotation, a method based on vector semantic matching driven by a large model is proposed. This method completes its design through problem analysis, appropriate modeling, summarizing from the large model, and matching with the labeling system. Additionally, agents assist in generating precise keywords during the RAG technology retrieval phase. Experiments have verified that this method achieves an accuracy exceeding 95% and described application scenarios. Looking forward, large language models can further optimize training strategies to better understand complex semantics and improve the accuracy of summary extraction. RAG technology can enhance the retrieval algorithms, improving retrieval speed and recall rate; agents can strengthen the analysis of text features and generate more targeted keywords. The three can further deepen cooperation and explore better integration mechanisms, such as developing a more intelligent interactive feedback system, enabling large language models to dynamically adjust based on RAG retrieval outcomes and agent analyses.

In the future, agents can design adaptive generation strategies through keyword design and online learning mechanisms to update keyword weight parameters to meet the long-term needs of domain knowledge evolving over time. Simultaneously, it could incorporate more advanced technologies, such as knowledge graphs, to provide richer information support for intelligent labeling methods, continually expanding application areas and enhancing the efficiency, accuracy, and generalization capability of text data labeling to meet the growing data processing demands across various industries.

References

[1] Naveed, H., Khan, A. U., Qiu, S., Saqib, M., Anwar, S., Usman, M., ... & Mian, A. (2023). A comprehensive overview of large language models. arxiv preprint arxiv:2307.06435.

[2] Chang, Y., Wang, X., Wang, J., Wu, Y., Yang, L., Zhu, K., ... & **e, X. (2024). A survey on evaluation of large language models. ACM transactions on intelligent systems and technology, 15(3), 1-45.

[3] Gao, Y., **ong, Y., Gao, X., Jia, K., Pan, J., Bi, Y., ... & Wang, H. (2023). Retrieval-augmented generation for large language models: A survey. arxiv preprint arxiv:2312.10997, 2.

[4] Jiang, Z., Xu, F. F., Gao, L., Sun, Z., Liu, Q., Dwivedi-Yu, J., ... & Neubig, G. (2023, December). Active retrieval augmented generation. In Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (pp. 7969-7992).

[5] Bang Liu, Xinfeng Li, Jiayi Zhang, Jinlin Wang, Tanjin He... & Chenglin Wu. (2025, Aarch). Advances and Challenges in Foundation Agents: From Brain-Inspired Intelligence to Evolutionary, Collaborative, and Safe Systems.Artificial Intelligence.arXiv:2504.01990.

[6] Emha, T. L., Yusoh, Z. I. M., & Aboobaider, B. M. (2022). BERT based named entity recognition for automated Hadith narrator identification. International Journal of Advanced Computer Science and Applications, 13(1).

[7] Sharaff, A., Chowdhury, T. R., & Bhandarkar, S. (2023). Lstm based sentiment analysis of financial news. SN Computer Science, 4(5), 584.

[8] Jo, S., Ryu, S., Kim, S., Yang, E., & Kim, K. (2024, September). TTD: Text-Tag Self-Distillation Enhancing Image-Text Alignment in CLIP to Alleviate Single Tag Bias. In European Conference on Computer Vision (pp. 341-357). Cham: Springer Nature Switzerland.

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